Question 10: Downloading and installing HBase; startup scripts and configuration files

and configuration files

Aim

To install Apache HBase, configure it to use Hadoop HDFS and start the HBase services.

Procedure

1. Install and configure a compatible Java version and Hadoop installation.

2. Download the binary distribution of Apache HBase and extract it, for example, to /usr/local/hbase.

3. Set HBASE_HOME and add its bin directory to PATH.

4. Configure hbase-env.sh with JAVA_HOME. Configure hbase-site.xml with the HDFS root directory and standalone or

distributed mode.

5. Ensure HDFS is running before starting HBase when HBase is configured to use HDFS.

6. Start HBase using start-hbase.sh and verify the master and region server processes.

7. Open the HBase web UI, commonly available on port 16010, if enabled.

Configuration

$HBASE_HOME/conf/hbase-env.sh

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

<!-- $HBASE_HOME/conf/hbase-site.xml -->

<configuration>

<property>

<name>hbase.rootdir</name>

<value>hdfs://localhost:9000/hbase</value>

</property>

<property>

<name>hbase.cluster.distributed</name>

<value>false</value>

</property>

<property>

<name>hbase.unsafe.stream.capability.enforce</name>

<value>false</value>

</property>

</configuration>

Startup scripts / commands

start-dfs.sh

start-hbase.sh

jps

Stop HBase:

stop-hbase.sh

Output of Program

$ jps

NameNode

DataNode

HMaster

HRegionServer

Jps

Result

HBase is installed and configured. The HBase master and region server are started and verified.

The example uses standalone HBase mode. For a fully distributed cluster, configure the appropriate distributed settings and

ZooKeeper services according to the installed HBase release.